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Title: A Comprehensive Survey of Explainable Artificial Intelligence Techniques for Malicious Insider Threat Detection

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Summary:

This paper provides an in-depth survey of explainable artificial intelligence (XAI) techniques applied to the detection of malicious insider threats, a critical issue for organizations dealing with internal security risks. The authors highlight the limitations of traditional detection methods, which often fail to identify covert insider activities that can compromise sensitive data and systems. By integrating XAI with machine learning (ML) and deep learning (DL) models, the study emphasizes the importance of interpretability to build trust among security analysts and improve detection accuracy. The survey reviews various XAI approaches, such as feature importance analysis, local interpretable model-agnostic explanations (LIME), and SHAP (SHapley Additive explanations), demonstrating how they enhance transparency in insider threat detection systems.

The methodology involves a systematic review of existing literature, categorizing XAI techniques based on their applicability to insider threat scenarios, including behavioral analysis, anomaly detection, and predictive modeling. Key findings reveal that XAI not only boosts detection rates by 15-25% in simulated environments but also addresses ethical concerns like bias mitigation and regulatory compliance. Case studies from sectors like finance and healthcare illustrate practical implementations, where XAI helps explain model decisions in real-time monitoring of employee activities, such as unusual data access patterns. Challenges discussed include computational overhead, data privacy under frameworks like GDPR, and the need for standardized benchmarks for XAI in threat detection.

In conclusion, the authors advocate for future research directions, including privacy-preserving AI, human-in-the-loop systems, and tailored benchmarking for insider threats. The paper positions XAI as essential for creating accountable and effective detection frameworks, particularly in high-stakes environments where insider threats pose significant financial and reputational risks. This work serves as a valuable resource for researchers and practitioners aiming to advance transparent AI solutions in cybersecurity.